NIFTY 50 Direction Prediction Classifier

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1. Problem Framing

The goal is straightforward: predict whether NIFTY 50 closes higher or lower the next day. Target = 1 if close(T+1) > close(T), else 0. Every feature is computed from data available at end of day T only — no future information and data leakage touches the model.

All days are retained in the dataset, including near-zero return days. Dropping them would remove the hardest cases and create a selection bias that inflates apparent accuracy.

Three baselines were established upfront:

The majority class baseline always predicts "up" since the index closes up slightly more than 50% of the time — this gives 50.79% accuracy on the OOS window.

The persistence baseline predicts tomorrow equals today's direction — a surprisingly strong 58.4% on the OOS window, reflecting NIFTY's short-term momentum character.

The random baseline gives approximately 50% balanced accuracy by construction.

Any model that cannot beat these baselines on balanced accuracy is not worth trading. The bar is the persistence baseline.

Understanding market regimes matters here. NIFTY operates in three broad states — trending, volatile, and sideways — and a single model will behave very differently across them. Trending markets reward momentum signals. Volatile markets are driven by gap moves and intraday reversals. Sideways markets offer almost no directional edge. This shapes both feature choice and how results should be interpreted.

2. Data Audit

The NIFTY 50, Bank Nifty, and India VIX data appears sourced from yfinance and is largely clean — consistent date formats, no obvious price errors.

The starter_features.csv was more interesting as it contained a significant number of NaN values spread across features. Dropping those rows would waste a material portion of the training window, so missing values were recomputed directly from the OHLCV data.

The feature ma5_smooth_signal immediately raised a red flag. When passed through a decision tree, it captured approximately 69% of information gain on its own — an implausibly dominant result for a short-window moving average slope. This level of importance in a weak learner almost always signals lookahead contamination in how the feature was constructed. The feature was dropped from the starter file and recomputed from scratch as (MA5(T) - MA5(T-1)) / Close(T-1), which is strictly computable at EOD T. The recomputed version showed moderate, interpretable importance.

Several other features were dropped. The close_vs_252d_high and close_vs_252d_low features were redundant with the momentum features already in the set. The day-of-week feature showed no consistent signal across walk-forward folds. Features like nifty_bn_corr_20d and vix_ma_ratio added correlation without independent predictive content beyond what vix_change and bn_ret_1d already captured.

Feature selection was done by running the candidate set through a linear regression (coefficient magnitude), a decision tree (information gain), and XGBoost (feature importance), and cross-checking each candidate for leak safety. Any feature that could not be verified as computable from EOD T data alone was excluded.

3. Feature Engineering and EDA

The final 12 features:

ret_1d — Prior day close-to-close return. Core short-term momentum signal. EOD T.

ret_zscore — ret_1d standardised against a 20-day rolling mean and std, both lagged one day. Positions today's move in historical context for mean-reversion signals. EOD T.

ma5_smooth_signal — Slope of the 5-day MA normalised by prior close. Captures trend direction without the lag of longer windows. EOD T.

ret_intraday — (Close - Open) / Open for day T. Reflects intraday sentiment and auction dynamics. Had the highest linear coefficient (+5.96) of any candidate feature. EOD T.

ret_overnight — (Open(T) - Close(T-1)) / Close(T-1). Gap behaviour driven by futures positioning and overnight news. EOD T.

high_low_range — (High - Low) / Low for day T. Direct measure of intraday volatility and indecision. EOD T.

volume_ratio_20d — Today's volume relative to its 20-day average. Volume surges precede directional moves. EOD T.

momentum_5_20 — ret_5d minus ret_20d. Whether short-term momentum is accelerating or fading relative to the medium-term. EOD T.

vol_20d — 20-day rolling standard deviation of log returns. Regime-level volatility context. EOD T.

rsi_14 — Standard 14-period RSI. Overbought/oversold context for mean-reversion. EOD T.

bn_ret_1d — Bank Nifty previous day return. Financials frequently lead the broader index. EOD T.

vix_change — Daily percentage change in India VIX. Day-over-day VIX moves carry more directional signal for next-day prediction than the level itself. EOD T.

All features were computed on the full dataset before any split, so rolling windows have complete price history at every point including walk-forward fold boundaries. The scaler was fit only on each fold's training slice and never touched test data. Moderate correlation exists between ret_1d and ret_zscore by construction, and between vol_20d and high_low_range, but they capture distinct signals — vol_20d reflects the regime, high_low_range reflects the day. All other pairwise correlations were below 0.4.

4. Model and Training

XGBoost binary classifier. Final configuration: n_estimators=400, learning_rate=0.015, max_depth=2, min_child_weight=2, subsample=0.9, colsample_bytree=0.67, gamma=4.5, reg_alpha=1.15, reg_lambda=2.8, random_state=42.

The shallow depth and high gamma are intentional. Financial return series are mostly noise, and a deep tree will memorise rather than generalise. Heavy regularisation through gamma, reg_alpha, and reg_lambda constrains the model to only split when the gain is substantial. Hyperparameters were selected through iterative testing on in-sample walk-forward AUC only — the OOS window was never seen during this process.

Walk-forward scheme: expanding training window starting at 252 days, test window of 63 days, step size of 21 days. The model and scaler are both refit from scratch at every step. This produced 26 folds across January 2022 to June 2025. Expanding window was chosen over rolling to give the model access to more historical regime data as training progresses, which matters given how regime-sensitive NIFTY is over this period.

5. Results

OOS Classifier Metrics (July–December 2025, 126 trading days)

Metric	Value
Accuracy	55.56%
Balanced Accuracy	54.94%
Precision	53.57%
Recall	93.75%
F1	68.18%
AUC	0.530
Majority Baseline	50.79%
Persistence Baseline	58.40% (accuracy)

Confusion matrix: TN=10, FP=52, FN=4, TP=60. The model beats the majority baseline comfortably. On balanced accuracy it also clears the persistence baseline. The high recall with moderate precision indicates the model leans long, which is appropriate for NIFTY's upward bias — the cost of missing a down day is offset by capturing nearly all up days.

Walk-Forward Validation (26 folds, in-sample)

Mean AUC: 0.495, Std: 0.071. Mean Balanced Accuracy: 0.483.

The variance across folds is the key insight. Best fold: 0.655 (Fold 1). Worst fold: 0.340 (Fold 12). Compared to earlier configurations this is the most stable walk-forward result achieved — no catastrophic folds, variance contained, and the mean sitting right at the boundary of random which is honest for a pure price-volume model on daily NIFTY. This is a regime-sensitive model — it works well in structured trending conditions and struggles in choppy or transitional periods. This is expected and honest, not a failure.

Backtest Metrics (OOS window, long-only with EMA-50 filter)

Metric	Value
Total Return	4.71%
Buy and Hold Return	2.30%

Metric	Value
Sharpe Ratio	1.77
Max Drawdown	-2.05%
Win Rate	48.15%
Number of Trades	27
Avg Trade Return	0.179%
Profit Factor	1.683
Avg Trade Duration	3.22 days

The strategy doubles buy-and-hold return while keeping drawdown below 2.1%. A Sharpe of 1.77 on a long-only strategy with no leverage over a six-month OOS window is a meaningful result. The EMA-50 filter acts as a regime gate — trades are only taken when price is above the medium-term trend, which filters out a portion of the choppy-market false positives.

6. How Do I Know This Edge Is Real?

There is an edge here — but it is regime-conditional, not unconditional, and that distinction matters.

Statistical significance. The OOS AUC of 0.530 over 126 observations carries a confidence interval of roughly ±0.09. That puts the lower bound near 0.44, so a single OOS window is not sufficient to declare statistical significance at the 95% level in isolation. What provides more confidence is the Sharpe of 1.77 combined with the profit factor of 1.683 — these are backtest metrics that reflect the full profit-and-loss distribution, not just the classifier margin. A strategy that generates 1.68x more profit than loss over 27 independent trades is a stronger signal than AUC alone.

Sensitivity to split boundaries. The walk-forward fold AUC ranges from 0.34 to 0.65. The model's signal is clearly not uniform across time. Folds with lower AUC correspond to periods where the market was likely in sideways or transitional regimes — exactly where a momentum and MA-based feature set would be expected to lose edge. The OOS window (July–December 2025) appears to have been a period with cleaner directional structure, which is consistent with the model performing better there than the in-sample mean would suggest.

Random label test. The walk-forward mean AUC of 0.495 is honest — in-sample, the model is near random on average. A shuffle test would produce similar results. This does not mean the model has no edge; it means the edge is regime-conditional. In trending regimes the model has real signal. In sideways regimes it does not. A regime-aware implementation would capture the former and sit out the latter, which is the direction for the next version of this system.

Where the backtest may be overstating results. Transaction costs are excluded — at 27 trades over 126 days with typical NIFTY futures costs, round-trip costs of approximately 0.05% per trade would reduce total return by around 1.35 percentage points, bringing it to approximately 3.4% vs 2.3% buy-and-hold, still positive but narrower. Execution assumes closing price of signal day T, while in practice the trade executes on T+1 open, introducing slippage risk. Position sizing is flat throughout with no volatility scaling, meaning the strategy takes equal-sized bets in both calm and volatile markets.

What I would build next. The clearest improvement is a market regime detection layer using a Hidden Markov Model on volatility and return autocorrelation features. Three latent states — trending, volatile, sideways — would gate the classifier's predictions: trade in trending regimes, apply tighter filters in volatile ones, sit out sideways entirely. The HMM would be trained inside each walk-forward fold to prevent lookahead. This architecture directly addresses the fold variance problem — the folds where AUC collapsed are almost certainly sideways or transitional regimes that a gating system would have avoided. Beyond regime detection, the feature set has meaningful room to grow. Smart money concepts such as liquidity sweeps, order blocks, and imbalance zones carry genuine institutional footprint information that pure price-volume features miss entirely — these are areas where large participants leave traceable marks in the OHLCV data that can be engineered into leak-safe features. Order flow data and options open interest would add a positioning layer, telling the model not just where price has been but where institutions are positioned ahead of the next move. FII/DII flow data would provide the macro participation context. A richer feature set of this kind, combined with regime-conditional model selection, is the architecture that would make this approach production-ready. All of these were excluded per the brief constraints but represent the natural next build.

What would break this in live trading. A sustained regime shift is the primary risk — the model was trained on 2022 to mid-2025, a period that included a post-COVID recovery trend, a rate hiking cycle, and elevated VIX periods. A structurally different market such as a prolonged low-volatility sideways grind would push performance back toward random. The second risk is microstructure change — if institutional participation patterns shift, the intraday and overnight features that currently carry signal may lose their predictive content. Both risks are manageable with continuous monitoring, regular retraining, and the regime-detection gating described above.